

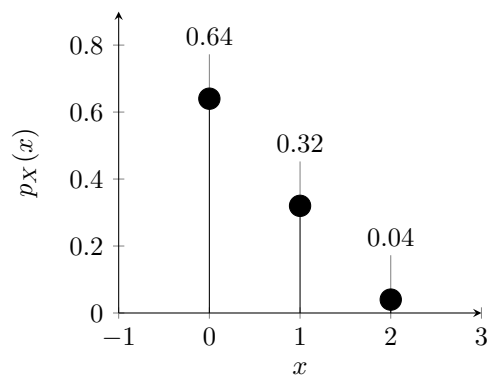
Solutions to Assignment 3

Q1 (10 points):

a) Since X is a Binomial random variable:

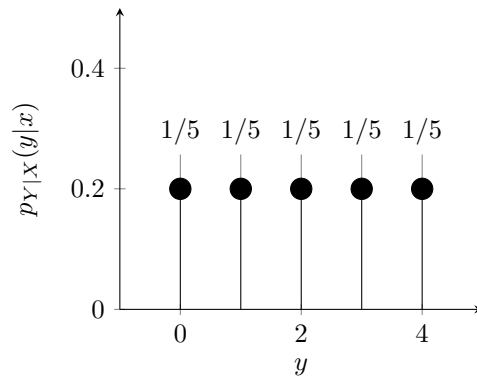
$$p_X(x) = {}^2C_x p^x (1-p)^{2-x}$$

$$= \begin{cases} {}^2C_0 0.2^0 (1-0.2)^2 = 0.64 & x = 0 \\ {}^2C_1 0.2^1 (1-0.2)^1 = 0.32 & x = 1 \\ {}^2C_2 0.2^2 (1-0.2)^0 = 0.04 & x = 2 \end{cases}$$

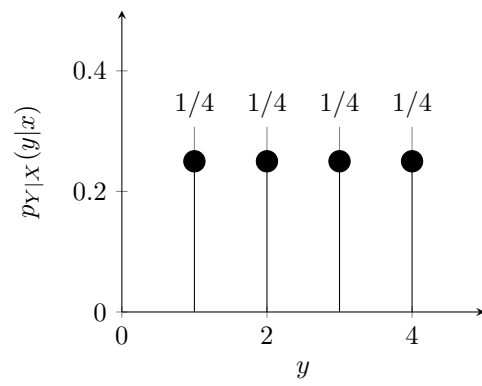


b)

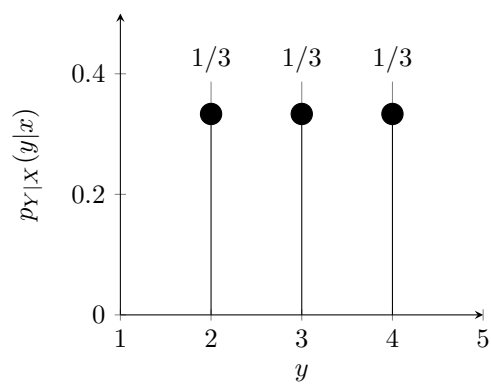
$$P_{Y|X}(y|0) = \begin{cases} \frac{1}{5} & y \in [0, 4] \\ 0 & \text{otherwise} \end{cases}$$



$$P_{Y|X}(y|1) = \begin{cases} \frac{1}{4} & y \in [1, 4] \\ 0 & \text{otherwise} \end{cases}$$

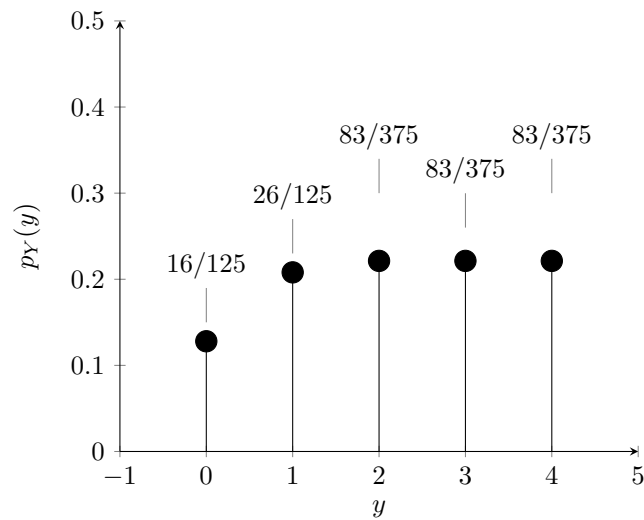


$$P_{Y|X}(y|2) = \begin{cases} \frac{1}{3} & y \in [2, 4] \\ 0 & \text{otherwise} \end{cases}$$



c) Using the total probability theorem:

$$\begin{aligned}
p_Y(y) &= \sum_x p_X(x) \cdot p_{Y|X}(y|x) \\
p_Y(0) &= \sum_x p_X(x) \cdot p_{Y|X}(0|x) \\
&= p_X(0) \cdot p_{Y|X}(0|0) + p_X(1) \cdot p_{Y|X}(0|1) + p_X(2) \cdot p_{Y|X}(0|2) \\
&= 0.64 \times \frac{1}{5} + 0.32 \times 0 + 0.04 \times 0 = \frac{16}{125} \\
p_Y(1) &= \sum_x p_X(x) \cdot p_{Y|X}(1|x) \\
&= p_X(0) \cdot p_{Y|X}(1|0) + p_X(1) \cdot p_{Y|X}(1|1) + p_X(2) \cdot p_{Y|X}(1|2) \\
&= 0.64 \times \frac{1}{5} + 0.32 \times \frac{1}{4} + 0.04 \times 0 = \frac{26}{125} \\
p_Y(2) &= \sum_x p_X(x) \cdot p_{Y|X}(2|x) \\
&= p_X(0) \cdot p_{Y|X}(2|0) + p_X(1) \cdot p_{Y|X}(2|1) + p_X(2) \cdot p_{Y|X}(2|2) \\
&= 0.64 \times \frac{1}{5} + 0.32 \times \frac{1}{4} + 0.04 \times \frac{1}{3} = \frac{83}{375} \\
p_Y(3) &= \sum_x p_X(x) \cdot p_{Y|X}(3|x) \\
&= p_X(0) \cdot p_{Y|X}(3|0) + p_X(1) \cdot p_{Y|X}(3|1) + p_X(2) \cdot p_{Y|X}(3|2) \\
&= 0.64 \times \frac{1}{5} + 0.32 \times \frac{1}{4} + 0.04 \times \frac{1}{3} = \frac{83}{375} \\
p_Y(4) &= \sum_x p_X(x) \cdot p_{Y|X}(4|x) \\
&= p_X(0) \cdot p_{Y|X}(4|0) + p_X(1) \cdot p_{Y|X}(4|1) + p_X(2) \cdot p_{Y|X}(4|2) \\
&= 0.64 \times \frac{1}{5} + 0.32 \times \frac{1}{4} + 0.04 \times \frac{1}{3} = \frac{83}{375} \\
p_Y(y) &= \begin{cases} \frac{16}{125} & y = 0 \\ \frac{26}{125} & y = 1 \\ \frac{83}{375} & y = 2 \\ \frac{83}{375} & y = 3 \\ \frac{83}{375} & y = 4 \end{cases}
\end{aligned}$$



d)

$$p_{X,Y}(x, y) = p_{Y|X}(y|x) \cdot p_X(x)$$

$$p_{X,Y}(0, 0) = p_{Y|X}(0|0) \cdot p_X(0) = \frac{1}{5} \times 0.64 = \frac{16}{125}$$

$$p_{X,Y}(0, 1) = p_{Y|X}(1|0) \cdot p_X(0) = \frac{1}{5} \times 0.64 = \frac{16}{125}$$

$$p_{X,Y}(0, 2) = p_{Y|X}(2|0) \cdot p_X(0) = \frac{1}{5} \times 0.64 = \frac{16}{125}$$

$$p_{X,Y}(0, 3) = p_{Y|X}(3|0) \cdot p_X(0) = \frac{1}{5} \times 0.64 = \frac{16}{125}$$

$$p_{X,Y}(0, 4) = p_{Y|X}(4|0) \cdot p_X(0) = \frac{1}{5} \times 0.64 = \frac{16}{125}$$

$$p_{X,Y}(1, 1) = p_{Y|X}(1|1) \cdot p_X(1) = \frac{1}{4} \times 0.32 = \frac{2}{25}$$

$$p_{X,Y}(1, 2) = p_{Y|X}(2|1) \cdot p_X(1) = \frac{1}{4} \times 0.32 = \frac{2}{25}$$

$$p_{X,Y}(1, 3) = p_{Y|X}(3|1) \cdot p_X(1) = \frac{1}{4} \times 0.32 = \frac{2}{25}$$

$$p_{X,Y}(1, 4) = p_{Y|X}(4|1) \cdot p_X(1) = \frac{1}{4} \times 0.32 = \frac{2}{25}$$

$$p_{X,Y}(2, 2) = p_{Y|X}(2|2) \cdot p_X(2) = \frac{1}{3} \times 0.04 = \frac{1}{75}$$

$$p_{X,Y}(2, 3) = p_{Y|X}(3|2) \cdot p_X(2) = \frac{1}{3} \times 0.04 = \frac{1}{75}$$

$$p_{X,Y}(2, 4) = p_{Y|X}(4|2) \cdot p_X(2) = \frac{1}{3} \times 0.04 = \frac{1}{75}$$

		X		
		0	1	2
Y	0	$\frac{16}{125}$	0	0
	1	$\frac{16}{125}$	$\frac{2}{25}$	0
	2	$\frac{16}{125}$	$\frac{2}{25}$	$\frac{1}{75}$
	3	$\frac{16}{125}$	$\frac{2}{25}$	$\frac{1}{75}$
	4	$\frac{16}{125}$	$\frac{2}{25}$	$\frac{1}{75}$

Table 1: $p_{X,Y}(x, y)$

Q2 (10 points):

Let H be the normal random variable that represents the height of the students. Then, we can standardize H using the linear transformation to obtain a standard normal variable N :

$$N = \frac{H - \mu}{\sigma}$$

Hence:

$$\begin{aligned} P(H \geq 72) &= P\left(N \geq \frac{H - \mu}{\sigma}\right) \\ &= P\left(N \geq \frac{72 - 65}{3}\right) \\ &= P(N \geq 2.33) \\ &= 1 - P(N \leq 2.33) \\ &= 1 - 0.9901 \\ &= 0.0099 \end{aligned}$$

Q3 (10 points):

a) $f(x)$ is a uniform random variable whose expected value is of the form:

$$E[X] = \frac{a+b}{2} = \frac{25+45}{2} = 35$$

b)

$$\begin{aligned} Var(X) &= E[(X - E[X])^2] \\ &= \int_{-\infty}^{\infty} (x - E[X])^2 \cdot f_X(x) dx \\ &= \int_a^b \left(x - \left(\frac{a+b}{2}\right)\right)^2 \cdot \frac{1}{20} dx \\ &= \frac{1}{20} \int_a^b \left(x - \left(\frac{a+b}{2}\right)\right)^2 dx \\ &= \frac{1}{20} \frac{\left(x - \left(\frac{a+b}{2}\right)\right)^3}{3(1)} \Big|_a^b \\ &= \frac{1}{60} \left(\left(b - \left(\frac{a+b}{2}\right)\right)^3 - \left(a - \left(\frac{a+b}{2}\right)\right)^3 \right) \\ &= \frac{1}{60} \left(\left(\frac{b-a}{2}\right)^3 - \left(\frac{a-b}{2}\right)^3 \right) \\ &= \frac{1}{60} \left(\left(\frac{b-a}{2}\right)^3 + \left(\frac{b-a}{2}\right)^3 \right) \\ &= \frac{1}{30} \left(\frac{b-a}{2}\right)^3 \\ &= \frac{1}{240} (b-a)^3 = \frac{1}{240} (20)^3 = \frac{100}{3} \end{aligned}$$

$$\sigma = \sqrt{Var(X)} = \sqrt{\frac{100}{3}} \approx 5.77$$

c)

$$\begin{aligned}P(25 \leq X \leq 30) &= \int_{25}^{30} f(x)dx \\&= \int_{25}^{30} \frac{1}{20}dx \\&= \frac{30 - 25}{20} \\&= \frac{5}{20} = \frac{1}{4}\end{aligned}$$

d)

$$\begin{aligned}P(30 \leq X \leq 40) &= \int_{30}^{40} f(x)dx \\&= \int_{30}^{40} \frac{1}{20}dx \\&= \frac{40 - 30}{20} \\&= \frac{10}{20} = \frac{1}{2}\end{aligned}$$

Q4 (10 points):

X = the number of kilometers you drive before getting into an accident. Since $f_X(x)$ is an exponential random variable, it is modeled as follows:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

For exponential random variables $E[X] = \frac{1}{\lambda} = \frac{1}{15000}$, hence $\lambda = \frac{1}{15000}$.

For exponential random variables:

$$\begin{aligned}P(X \geq a) &= e^{-\lambda a} \\P(X \geq 4200) &= e^{-\frac{1}{15000} \times 4200} \\&= e^{-\frac{7}{25}} \approx 0.756\end{aligned}$$

Q5 (10 points):

a) The following strategy can be used:

- Calculate the CDF of X
- Use the CDF to find the PDF of X

Let X_A , X_B , and X_C be the random variables that represent the distance the javelin is thrown by Arshad, Babar, and Shaheen respectively. Since all distances the javelin is thrown are uniformly distributed within the respective distance brackets for each athlete:

$$f_{X_A}(x) = \begin{cases} \frac{1}{10} & 80 \leq x \leq 90 \\ 0 & \text{otherwise} \end{cases}$$

$$f_{X_B}(x) = \begin{cases} \frac{1}{5} & 80 \leq x \leq 85 \\ 0 & \text{otherwise} \end{cases}$$

$$f_{X_C}(x) = \begin{cases} \frac{1}{10} & 75 \leq x \leq 85 \\ 0 & \text{otherwise} \end{cases}$$

The CDF of X is given by:

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P(X_A \leq x, X_B \leq x, X_C \leq x) \end{aligned}$$

Since the throws of each athlete are independent of the other:

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P(X_A \leq x) \cdot P(X_B \leq x) \cdot P(X_C \leq x) \\ &= F_{X_A}(x) \cdot F_{X_B}(x) \cdot F_{X_C}(x) \end{aligned}$$

$$F_{X_A}(x) = \begin{cases} 0 & x \leq 80 \\ \frac{x-80}{10} & 80 \leq x \leq 90 \\ 1 & x \geq 90 \end{cases}$$

$$F_{X_B}(x) = \begin{cases} 0 & x \leq 85 \\ \frac{x-85}{5} & 85 \leq x \leq 90 \\ 1 & x \geq 90 \end{cases}$$

$$F_{X_C}(x) = \begin{cases} 0 & x \leq 75 \\ \frac{x-75}{10} & 75 \leq x \leq 85 \\ 1 & x \geq 85 \end{cases}$$

Hence:

$$\begin{aligned} F_X(x) &= \begin{cases} 0 & x \leq 85 \\ \left(\frac{x-80}{10}\right) \left(\frac{x-85}{5}\right) (1) & 85 \leq x \leq 90 \\ 1 & x \geq 90 \end{cases} \\ &= \begin{cases} 0 & x \leq 85 \\ \frac{(x-80)(x-85)}{50} & 85 \leq x \leq 90 \\ 1 & x \geq 90 \end{cases} \\ &= \begin{cases} 0 & x \leq 85 \\ \frac{(x^2 - 165x + 6800)}{50} & 85 \leq x \leq 90 \\ 1 & x \geq 90 \end{cases} \end{aligned}$$

$$f_X(x) = \frac{d(F_X(x))}{dx}$$

In the range $85 \leq x \leq 90$:

$$\begin{aligned} f_X(x) &= \frac{d}{dx} \left(\frac{x^2 - 165x + 6800}{50} \right) \\ f_X(x) &= \frac{2x - 165}{50} \end{aligned}$$

Finally:

$$f_X(x) = \begin{cases} \frac{2x - 165}{50} & 85 \leq x \leq 90 \\ 0 & \text{otherwise} \end{cases}$$

b i)

$$f_X(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \begin{cases} 1 & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Since X and Y are independent:

$$f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$$

$$= \begin{cases} 0.5 & 0 \leq x \leq 2, \quad 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f_{X,Y}(1.5, 1.5) = 0$$

b ii)

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

$$= F_X(x) \cdot F_Y(y)$$

$$= \begin{cases} \frac{xy}{2} & 0 \leq x \leq 2, 0 \leq y \leq 1 \\ y & x \geq 2, \quad 0 \leq y \leq 1 \\ \frac{x}{2} & 0 \leq x \leq 2, \quad y \geq 1 \\ 1 & x \geq 2, \quad y \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

b iii)

$$f_{X|A} = \begin{cases} \frac{f_X(x)}{P(X \in A)} & \text{if } x \in A \\ 0, & \text{otherwise} \end{cases}$$

$$P(X \in A) = P(0.25 \leq x \leq 0.5)$$

$$= \frac{0.5 - 0.25}{2} = 0.125$$

$$f_{X|A} = \begin{cases} \frac{0.5}{0.125} = 4 & \text{if } x \in A \\ 0, & \text{otherwise} \end{cases}$$

$$E[X|A] = \frac{a+b}{2}$$

$$= \frac{0.25 + 0.5}{2} = \frac{0.75}{2} = 0.375$$

Q6 (10 points):

a)

$$f_{X,Y}(x,y) = \begin{cases} \frac{xy^2}{81} & 0 \leq x \leq 3, \quad 0 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

$$= \int_0^3 \frac{xy^2}{81} dy$$

$$= \frac{x}{81} \cdot \frac{y^3}{3} \Big|_0^3$$

$$= \frac{x}{81} \left(\frac{3^3}{3} \right)$$

$$= \frac{x}{9}$$

$$f_X(x) = \begin{cases} \frac{x}{9} & , \quad 0 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

$$= \int_0^3 \frac{xy^2}{81} dy$$

$$= \frac{y^2}{81} \cdot \frac{x^2}{2} \Big|_0^3$$

$$= \frac{y^2}{81} \left(\frac{3^2}{2} \right)$$

$$= \frac{y^2}{18}$$

$$f_Y(y) = \begin{cases} \frac{y^2}{18} & , \quad 0 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} f_X(x) \cdot f_Y(y) &= \frac{x}{9} \cdot \frac{y^2}{18} \\ &= \frac{xy^2}{162} \end{aligned}$$

Since $f_{X,Y}(x,y) \neq f_X(x) \cdot f_Y(y)$, X and Y are not independent.

b i)

$$f_{X,Y}(x,y) = \begin{cases} x + cy^3 & , \quad 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned}
\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx dy &= 1 \\
\int_0^1 \int_0^1 x + cy^3 dx dy &= 1 \\
\int_0^1 \left. \frac{x^2}{2} + cxy^3 \right|_0^1 dy &= 1 \\
\int_0^1 \frac{1}{2} + cy^3 dy &= 1 \\
\left. \frac{1}{2}y + \frac{cy^4}{4} \right|_0^1 &= 1 \\
\frac{1}{2} + \frac{c}{4} &= 1 \\
c &= 2
\end{aligned}$$

b ii)

$$\begin{aligned}
f_{X,Y}(x,y) &= \begin{cases} x + 2y^3 & , \quad 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases} \\
f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy \\
f_X(x) &= \int_0^1 x + 2y^3 dy \\
f_X(x) &= xy + \frac{y^4}{2} \Big|_0^1 \\
f_X(x) &= x + \frac{1}{2} \\
f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx \\
f_Y(y) &= \int_0^1 x + 2y^3 dx \\
f_Y(y) &= \frac{x^2}{2} + 2xy^3 \Big|_0^1 \\
f_Y(y) &= \frac{1}{2} + 2y^3
\end{aligned}$$

Since $f_{X,Y}(x,y) \neq f_X(x) \cdot f_Y(y)$, X and Y are not independent.

Q7 (10 points):

i)

$$\begin{aligned}
f_{X,Y}(x,y) &= \begin{cases} \frac{2xy^2}{81} & 0 \leq x \leq 3, \quad 0 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases} \\
P(X \leq 1, Y \leq 1) &= \int_{-\infty}^1 \int_{-\infty}^1 f_{X,Y}(x,y) dx dy \\
&= \int_0^1 \int_0^1 \frac{2xy^2}{81} dx dy \\
&= \int_0^1 \frac{2y^2}{81} \cdot \frac{x^2}{2} \Big|_0^1 dy \\
&= \int_0^1 \frac{y^2}{81} dy \\
&= \frac{y^3}{243} \Big|_0^1 \\
&= \frac{1}{243}
\end{aligned}$$

ii)

$$\begin{aligned}
f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy \\
f_X(x) &= \int_0^3 \frac{2xy^2}{81} dy \\
f_X(x) &= \frac{2x}{81} \cdot \frac{y^3}{3} \Big|_0^3 \\
f_X(x) &= \frac{2x}{9} \\
f_X(x) &= \begin{cases} \frac{2x}{9} & 0 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases} \\
f_{X|A}(x) &= \begin{cases} \frac{f_X(x)}{P(A)} & x \in A \\ 0 & \text{otherwise} \end{cases} \\
P(A) &= P(0 \leq x \leq 1) = \int_0^1 \frac{2x}{9} dx \\
&= \frac{2}{9} \cdot \frac{x^2}{2} \Big|_0^1 \\
&= \frac{1}{9} \\
f_{X|A}(x) &= \begin{cases} \frac{\frac{2x}{9}}{\frac{1}{9}} & x \in A \\ 0 & \text{otherwise} \end{cases} \\
f_{X|A}(x) &= \begin{cases} 2x & x \in A \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

$$\begin{aligned}
E[X|A] &= \int_{-\infty}^{\infty} x \cdot f_{X|A}(x) dx \\
&= \int_0^1 x \cdot 2x dx \\
&= 2 \cdot \frac{x^3}{3} \Big|_0^1 \\
&= \frac{2}{3}
\end{aligned}$$

Q8 (10 points):

a)

$$\begin{aligned}
f_X(x) &= \begin{cases} 0.5 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases} \\
f_Y(y) &= \begin{cases} 0.5 & 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

Since both X and Y are independent:

$$\begin{aligned}
f_{X,Y}(x,y) &= f_X(x) \cdot f_Y(y) \\
&= \begin{cases} 0.25 & 0 \leq x \leq 2 \quad 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

b)

$$\begin{aligned}
F_X(x) &= \begin{cases} 0 & x \leq 0 \\ 0.5x & 0 \leq x \leq 2 \\ 1 & x \geq 2 \end{cases} \\
F_Y(y) &= \begin{cases} 0 & y \leq 0 \\ 0.5y & 0 \leq y \leq 2 \\ 1 & y \geq 2 \end{cases} \\
F_{X,Y}(1.5, 1.5) &= P(X \leq 1.5, Y \leq 1.5) \\
&= F_X(y \leq 1.5) \cdot F_Y(y \leq 1.5) \\
&= 0.5(1.5) \cdot 0.5(1.5) \\
&= 0.5625
\end{aligned}$$

c) Since X and Y are independent, the following holds true:

$$\begin{aligned}
f_{X,Y}(x,y) &= f_X(x) \cdot f_Y(y) \quad \forall \quad x, y, \text{ and} \\
f_{X,Y}(x,y) &= f_{Y|X}(y|x) \cdot f_X(x) \quad \forall \quad x \text{ and } y \text{ with } f(x) > 0
\end{aligned}$$

By comparing the two, we can see that:

$$\begin{aligned}
f_{Y|X}(y|x) &= f_Y(y) \\
f_{Y|X}(y|0.5) &= f_Y(y) \\
&= \begin{cases} 0.5 & 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

d) By the same argument as in part (c):

$$\begin{aligned}
f_{X|Y}(x|y) &= f_X(x) \\
f_{X|Y}(x|y) &= f_X(x) \\
&= \begin{cases} 0.5 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases} \\
f_{X|Y}(0.5|1) &= 0.5
\end{aligned}$$

e) Let A be the event $X|Y > 1.5$.

Since X and Y are independent:

$$\begin{aligned}
 f_{X|A}(x) &= f_X(x) \\
 E[X|A] &= \int_{-\infty}^{\infty} x \cdot f_{X|A}(x) dx \\
 &= \int_{-\infty}^{\infty} x \cdot f_X(x) dx \\
 &= \int_0^2 x \cdot 0.5 dx \\
 &= 0.5 \cdot \left. \frac{x^2}{2} \right|_0^2 \\
 &= 0.5 \cdot 2 = 1
 \end{aligned}$$

Q9 (10 points):

a) N is a standard normal random variable because it's mean is 0 and it has a unit variance.

$$\begin{aligned}
 A &= N \in [-0.25, 0.25] \\
 f_{N|A}(n) &= \begin{cases} \frac{f_N(n)}{P(A)} & -0.25 \leq N \leq 0.25 \\ 0 & \text{otherwise} \end{cases}
 \end{aligned}$$

Since $f_{N|A}$ is also normally distributed, it must be symmetrical in the interval $[-0.25, 0.25]$.
Hence the expected value $E(N|A) = 0$

b) Let X = The odometer reading of a randomly selected car in Karachi.

Let Y = The color of a randomly selected car in Karachi.

Then:

$$E[X|Y] = 55000$$

Represents the information that the average odometer reading of a randomly selected car in Karachi is 55000km given that it is red.

Q10 (10 points):

Let A be the event that the match is played in Asia. Then:

$$\begin{aligned}
 f_{X|A}(x) &= \begin{cases} 0.1 & 30.5 \leq x \leq 40.5 \\ 0 & \text{otherwise} \end{cases} \\
 f_{X|A^C}(x) &= \begin{cases} 0.05 & 15.5 \leq x \leq 30.5 \\ 0 & \text{otherwise} \end{cases} \\
 P(A) &= 0.6 \\
 P(A^C) &= 0.4
 \end{aligned}$$

By the total expectation theorem:

$$\begin{aligned}
 f_X(x) &= P(A)f_{X|A}(x) + P(A^C)f_{X|A^C}(x) \\
 &= \begin{cases} (0.6)(0) + (0.4)(0.05) & 15.5 \leq x \leq 30.5 \\ (0.6)(0.1) + (0.4)(0.05) & 30.5 \leq x \leq 35.5 \\ (0.6)(0.1) + (0.4)(0) & 30.5 \leq x \leq 35.5 \\ 0 & \text{otherwise} \end{cases}
 \end{aligned}$$

Q11 (10 points):

Let I be the discrete random variable that takes on the following values:

$I = 0$ When the ISP is Charter

$I = 1$ When the ISP is Xfinity

$$p_I(i) = \begin{cases} 0.3 & I = 0 \\ 0.7 & I = 1 \end{cases}$$

Let S be the internet speed experienced by a customer. Then:

$$f_{S|I}(s|0) = \begin{cases} \frac{1}{8} & 7 \leq s \leq 15 \\ 0 & \text{otherwise} \end{cases}$$
$$f_{S|I}(s|1) = \begin{cases} \frac{1}{10} & 10 \leq s \leq 20 \\ 0 & \text{otherwise} \end{cases}$$

We require $f_{I|S}(0|12)$.

By the total probability theorem:

$$f_S(s) = \sum_i p_I(i) f_{S|I}(s|i)$$
$$= p_I(0) f_{S|I}(s|0) + p_I(1) f_{S|I}(s|1)$$

Using Bayes' Rule:

$$p_{I|S}(i|s) = \frac{p_I(i) f_{S|I}(s|i)}{f_S(s)}$$
$$p_{I|S}(0|12) = \frac{p_I(0) f_{S|I}(12|0)}{f_S(12)}$$
$$p_{I|S}(0|12) = \frac{p_I(0) f_{S|I}(12|0)}{p_I(0) f_{S|I}(12|0) + p_I(1) f_{S|I}(12|1)}$$
$$p_{I|S}(0|12) = \frac{0.3 \times \frac{1}{8}}{0.3 \times \frac{1}{8} + 0.7 \times \frac{1}{10}}$$
$$\approx 0.349$$

Q12 (10 points):

$$p_S(s) = \begin{cases} 0.4 & s = 0 \\ 0.6 & s = 1 \end{cases}$$

$$f_{Y|S}(y|1) = \frac{1}{\sqrt{2\pi}} e^{\frac{-(y-1)^2}{2}}$$

$$f_{Y|S}(y|0) = \frac{1}{\sqrt{2\pi}} e^{\frac{-y^2}{2}}$$

By the total probability theorem:

$$f_Y(y) = \sum_s p_S(s) f_{Y|S}(y|s)$$
$$= p_S(0) f_{Y|S}(y|0) + p_S(1) f_{Y|S}(y|1)$$
$$= \frac{0.4}{\sqrt{2\pi}} e^{\frac{-y^2}{2}} + \frac{0.6}{\sqrt{2\pi}} e^{\frac{-(y-1)^2}{2}}$$

Using Bayes' Rule:

$$\begin{aligned}
 p_{S|Y}(s|y) &= \frac{p_S(s)f_{Y|S}(y|s)}{f_Y(y)} \\
 p_{S|Y}(1|0.7) &= \frac{p_S(1)f_{Y|S}(0.7|1)}{f_Y(0.7)} \\
 &= \frac{0.6 \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{(0.7-1)^2}{2}} \right)}{\frac{0.4}{\sqrt{2\pi}} e^{-\frac{0.7^2}{2}} + \frac{0.6}{\sqrt{2\pi}} e^{-\frac{(0.7-1)^2}{2}}} \\
 &= \frac{0.6e^{-0.045}}{0.4e^{-0.245} + 0.6e^{-0.045}} \\
 &= \frac{0.5736}{0.3131 + 0.5736} \\
 &= 0.647
 \end{aligned}$$

Q13 (10 points):

- a) Let X be the random variable that takes the value 0 when it's a sunny day and 1 when it's a rainy day.

$$\begin{aligned}
 p_X(x) &= \begin{cases} 0.75 & \text{when } x = 0 \\ 0.25 & \text{when } x = 1 \end{cases} \\
 f_{T|X}(t|0) &= \begin{cases} \frac{1}{15} & 30 \leq t \leq 45 \\ 0 & \text{otherwise} \end{cases} \\
 f_{T|X}(t|1) &= \begin{cases} \frac{1}{20} & 40 \leq t \leq 60 \\ 0 & \text{otherwise} \end{cases} \\
 f_T(t) &= \sum_x p_X(x)f_{T|X}(t|x) \\
 &= p_X(0)f_{T|X}(t|0) + p_X(1)f_{T|X}(t|1) \\
 &= \begin{cases} 0.05 & 30 \leq t \leq 40 \\ 0.0625 & 40 \leq t \leq 45 \\ 0.0125 & 45 \leq t \leq 60 \end{cases}
 \end{aligned}$$

- b) Using Bayes' Rule:

$$\begin{aligned}
 p_{X|T}(x|t) &= \frac{p_X(x) \cdot f_{T|X}(t|x)}{f_T(t)} \\
 p_{X|T}(1|42) &= \frac{p_X(1) \cdot f_{T|X}(42|1)}{f_T(42)} \\
 &= \frac{0.25 \times 0.05}{0.0625} \\
 &= 0.2
 \end{aligned}$$

Q14 (10 points):

We can approach this question using the general version of Bayes' Rule:

Prior Belief:

$$p_X(x) = \begin{cases} 0.1 & x = 1 \\ 0.9 & x = 0 \end{cases}$$

Quality of measurement:

$$f_{Y|X}(y|0) = \begin{cases} \frac{1}{4} & -2 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$
$$f_{Y|X}(y|1) = \begin{cases} \frac{1}{4} & -1 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Posterior Belief:

$$\begin{aligned} p_{X|Y}(x|y) &= \frac{p_X(x) \cdot f_{Y|X}(y|x)}{f_Y(y)} \\ &= \frac{p_X(x) \cdot f_{Y|X}(y|x)}{p_X(0)f_{Y|X}(y|0) + p_X(1)f_{Y|X}(y|1)} \\ p_{X|Y}(0|0.8) &= \frac{p_X(0) \cdot f_{Y|X}(0.8|0)}{p_X(0)f_{Y|X}(0.8|0) + p_X(1)f_{Y|X}(0.8|1)} \\ &= \frac{0.9 \cdot 0.25}{0.9 \times 0.25 + 0.1 \times 0.25} \\ &= 0.9 \\ p_{X|Y}(1|0.8) &= \frac{p_X(1) \cdot f_{Y|X}(0.8|1)}{p_X(0)f_{Y|X}(0.8|0) + p_X(1)f_{Y|X}(0.8|1)} \\ &= \frac{0.1 \cdot 0.25}{0.9 \times 0.25 + 0.1 \times 0.25} \\ &= 0.1 \end{aligned}$$

Comparing the values of $p_{X|Y}(0|0.8)$ and $p_{X|Y}(1|0.8)$, if the observation is 0.8, the value of X is most likely 0.

	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.5279	0.53188	0.53586
0.1	0.53983	0.5438	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.6293	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.6591	0.66276	0.6664	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.7054	0.70884	0.71226	0.71566	0.71904	0.7224
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.7549
0.7	0.75804	0.76115	0.76424	0.7673	0.77035	0.77337	0.77637	0.77935	0.7823	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.8665	0.86864	0.87076	0.87286	0.87493	0.87698	0.879	0.881	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.9032	0.9049	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.9222	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.9452	0.9463	0.94738	0.94845	0.9495	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.9608	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.9732	0.97381	0.97441	0.975	0.97558	0.97615	0.9767
2	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.9803	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.983	0.98341	0.98382	0.98422	0.98461	0.985	0.98537	0.98574
2.2	0.9861	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.9884	0.9887	0.98899
2.3	0.98928	0.98956	0.98983	0.9901	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.9918	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.9943	0.99446	0.99461	0.99477	0.99492	0.99506	0.9952
2.6	0.99534	0.99547	0.9956	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.9972	0.99728	0.99736
2.8	0.99744	0.99752	0.9976	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.999
3.1	0.99903	0.99906	0.9991	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.9994	0.99942	0.99944	0.99946	0.99948	0.9995
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.9996	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.9997	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.9998	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.9999	0.9999	0.9999	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997
4	0.99997	0.99997	0.99997	0.99997	0.99997	0.99997	0.99998	0.99998	0.99998	0.99998

Table 2: The standard normal table